Project Executability

Scientific Calculator (Java Swing)

Main class: ScientificCalculator
Type: Standalone desktop application. It needs no database, internet connection or external libraries.

1. System Requirements

Item	Requirement
Operating system	Windows, macOS or Linux
Java	JDK 8 or higher
Screen	1000 x 720 or larger recommended (the window opens at 1000 x 720)
External libraries	None (Swing and AWT come with the JDK)

To check that Java is installed, run these in a terminal. Both should print a version number:

```
java -version
javac -version
```

2. Running from the Command Line

- Save the code as **ScientificCalculator.java**. The file name must match the class name exactly.
- Open a terminal in that folder.
- Compile and run the program:

```
javac ScientificCalculator.java
java ScientificCalculator
```

Compiling creates a ScientificCalculator.class file. After you run it, the calculator window opens with 0 on the display.

3. Running from an IDE

- **IntelliJ IDEA / Eclipse / NetBeans:** Create a new Java project, add ScientificCalculator.java to the src folder, then right-click the file and choose Run.
- **VS Code:** Install the Extension Pack for Java, open the file, and click Run above the main method.

4. Verification Test Cases

After launching, confirm that everything works with these checks:

Input	Expected output
2 + 3 * 4 =	14
(2 + 3) * 4 =	20

Input	Expected output
2 y^x 3 =	8
5 then x!	120
16 then sqrt button	4
1000 then log	3
10 then 2^x	1024
4 then 1/x	0.25
50 then %, then =	0.5
Press Rad (it switches to Deg), then 90 and sin	1
5 / 0 =	Error
C	0

5. Common Problems and Fixes

Problem	Cause	Fix
'javac' is not recognized	JDK is not installed or not in PATH	Install the JDK and add its bin folder to PATH
class ScientificCalculator is public, should be declared in a file named...	File name does not match the class name	Rename the file to ScientificCalculator.java
Could not find or load main class	Wrong folder, or the program was not compiled first	Run java in the same folder as the .class file
Special symbols show as boxes	Source file is not saved as UTF-8	Compile with: javac -encoding UTF-8 ScientificCalculator.java
Button colors look different	The system look-and-feel overrides them	Cosmetic only; the calculator still works